WEEK 2 · LESSON 8 OF 24 · HANDOUT

The Machine-Learning Workflow

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=nKW8Ndu7Mjw — open the lesson online to watch inside the course.

Your goal in this lesson: List the stages of a machine-learning project in order, explain what happens at each stage, and see why the workflow is a loop, not a straight line.

From idea to working model

Real machine-learning projects follow a repeatable path. Knowing it helps you plan, estimate time and talk clearly with engineers, managers and clients.

The seven stages

#	Stage	Key question	Example: predicting maize yield for farmers
1	**Define the problem**	What decision will the prediction support, and how do we measure success?	Help a cooperative plan storage; success = error under 10%
2	**Collect data**	What data exists, who owns it, and may we use it?	Past yields, rainfall, soil type, fertiliser use
3	**Prepare the data**	Is it clean, complete and representative?	Fix missing rainfall, unify district names
4	**Choose and train a model**	Which method fits the problem type?	A regression model, because yield is a number
5	**Evaluate**	How well does it work on data it has never seen?	Test on last season's farms
6	**Deploy**	How will people actually use it?	A simple SMS or web dashboard for the cooperative
7	**Monitor and improve**	Is it still accurate as the world changes?	Check each season; retrain with new data

Stage 1 is the most important

Many failed projects had good models but a **badly defined problem**. Ask:

- Who will use the prediction, and what will they do differently?
- What does "good enough" mean in numbers?
- What would a simple non-AI solution achieve? (Sometimes a spreadsheet rule is enough.)

Feature engineering

During preparation, teams often create better features from raw data. From a date you can create *month* and *season*. From total sales and number of days you can create *average daily sales*. Good features can improve a simple model more than switching to a complex one.

Train, then test on new data

A model must be tested on examples it did **not** learn from — otherwise you only measure memory, not real skill. You will study this deeply in Week 3 (training, validation and test sets).

Deployment is not the end

The world changes: prices rise, weather patterns shift, customers change habits. A model trained on old data slowly becomes less accurate. This is called **model drift**. That is why stage 7 sends the team back to stages 2–5: the workflow is a **loop**.

Who does what

- **Domain experts** (farmers, doctors, accountants) — define the problem and judge whether results make sense.
- **Data engineers** — collect, store and move data.
- **Data scientists / ML engineers** — prepare data, train and evaluate models.
- **Software engineers** — build the app or system people use.
- **Product managers** — keep everyone focused on the user's real need.

Remember this: Define → collect → prepare → train → evaluate → deploy → monitor, and then loop back. A clear problem definition and good data matter more than a fancy model.

Name: _____ Date: _____

Activity

Task 1

Pick one problem from your community (for example: predicting which patients will miss clinic appointments, or which products a shop will run out of). Write a one-page plan with a short paragraph for each of the seven stages. Mark which stage you think will be hardest, and why.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is the first stage of a machine-learning project?
   - A. Train the model
   - B. Buy a GPU
   - C. Define the problem and how success is measured
   - D. Deploy the app
2. Predicting maize yield in tonnes is which problem type?
   - A. Regression
   - B. Generation
   - C. Clustering
   - D. Classification
3. Why must a model be tested on data it did not learn from?
   - A. To save electricity
   - B. Because training data is secret
   - C. It does not need testing
   - D. To measure real skill rather than memory
4. Creating 'average daily sales' from total sales and days is an example of…
   - A. Deployment
   - B. Data leakage
   - C. Model drift
   - D. Feature engineering
5. What is model drift?
   - A. A type of neural network
   - B. When training finishes early
   - C. When a model becomes less accurate because the world has changed
   - D. When a model moves to a new server
6. Why is the ML workflow described as a loop?
   - A. Because deployment comes first
   - B. Because the steps are random
   - C. Because models are trained once and never touched
   - D. Because monitoring leads back to new data, preparation and retraining

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